

First order Circuits :-

An Introduction to RC and RL circuits. Applying KCL, KVL

Produces differential equations.

First order circuit is characterized by a first order differential equation.

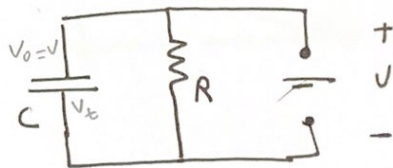
There are two ways to excite the circuits

- 1- Source-free circuits (stored energy causes current to flow in the circuit).
- 2- With independent source.

Source-Free RC Circuit

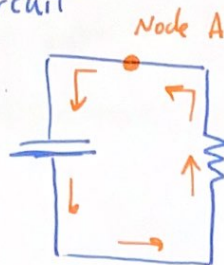
It ~~occurs~~ occurs when its dc source is suddenly disconnected.

The objective is to determine the circuit Response, which in this case the voltage $V_c(t)$ across the capacitor, when we disconnect the DC source.



- 1- The capacitor is Fully charged at $t=0 \Rightarrow V_c(0) = V_0 = V$
- 2- current will start to flow in the circuit
- 3- Apply KCL at the node A

$$i_c + i_R = 0$$



As we know $i_c = C \frac{dv}{dt}$, $i_R = \frac{V}{R}$

$$C \frac{dv}{dt} + \frac{V}{R} = 0 \quad \text{or} \quad \underbrace{\frac{dv}{dt} + \frac{V}{RC}}_{\downarrow} = 0$$

to solve it:

$$\frac{dv}{V} = -\frac{1}{RC} dt \quad (\text{Integrating})$$

$$\dot{V} + \frac{1}{RC} V = 0$$

First-order D.E.

$$\ln v = -\frac{t}{RC} + \ln A \Rightarrow \ln \frac{V}{A} = -\frac{t}{RC} \quad (\text{taking power of } e)$$

$$V_{(t)} = A e^{-t/RC} \quad \left| \quad \text{or} \quad \int_{V_0}^V \frac{dv}{V} = -\frac{1}{RC} \int_0^t dt \Rightarrow \ln v \Big|_{V_0}^V = -\frac{1}{RC} t \Big|_0^t \right.$$

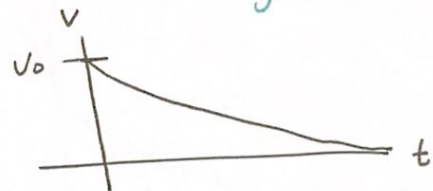
$$\ln v - \ln V_0 = -\frac{t}{RC} \Rightarrow \ln \frac{V}{V_0} = -\frac{t}{RC} \Rightarrow V = V_0 e^{-t/RC}$$

As in the initial conditions, $V_{(0)} = V_0$ at $t=0 \Rightarrow A = V_0$

$$V_{(t)} = V_0 e^{-t/RC}$$

The voltage response of the RC circuit is an exponential decay of the initial voltage.

This response is called Natural Respose, as it due to the initial energy stored not from external voltage or current source.



When $t = RC \Rightarrow V_t = V_0 e^{-1} \Rightarrow V_t = 0.368 V_0$

at this time $t = RC$ the voltage response decrease by 36% of its initial value.

This time called Time Response or τ

$$\tau = RC$$

At $t = RC$ or $t = \tau \Rightarrow \frac{V_t}{V_0} = 0.367$

t	$\frac{V_t}{V_0}$	
τ	0.367	36%
2τ	0.135	13%
3τ	0.049	4%
5τ	0.0067	0.6% after 5τ (Five time constant)

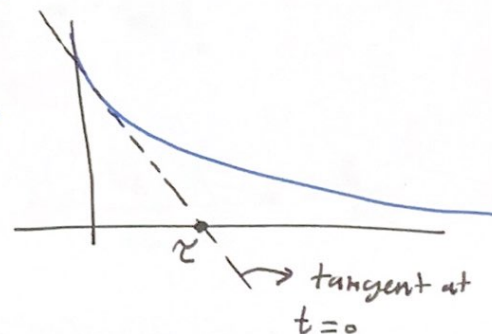
Small time constant $\Rightarrow$ Faster response
quick dissipation of energy.

Large time constant $\Rightarrow$ Slow response

At any rate, whether the time constant is small or large, the circuit reaches steady state in 5τ .

The power dissipated in the resistor is

$$P_R(t) = V i_R = \frac{V_0^2}{R} e^{-2t/\tau}$$



The energy absorbed by the resistor up to time t

$$W_{R(t)} = \int_0^t P_R d\tau = \int_0^t \frac{V_0^2}{R} e^{-2\tau/\tau} d\tau$$

$$= \frac{1}{2} C V_0^2 (1 - e^{-2t/\tau})$$

$$\left. \frac{d}{dt} \left(\frac{V}{V_0} \right) \right|_{t=0} = -\frac{1}{RC} e^{-t/RC} = -\frac{1}{RC} = -\frac{1}{\tau}$$

$$i_R = \frac{V_t}{R} = \frac{V_0}{R} e^{-t/\tau}$$

$t \rightarrow \infty$ $W_{R(\infty)} = W_C(0)$ the energy initially stored in the capacitor.